

Batman (Continued)

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1 Opener

Got a Batman finally! I used $\theta = [1.5, 0.4, 1.0, 0.2]$ and got this graph.

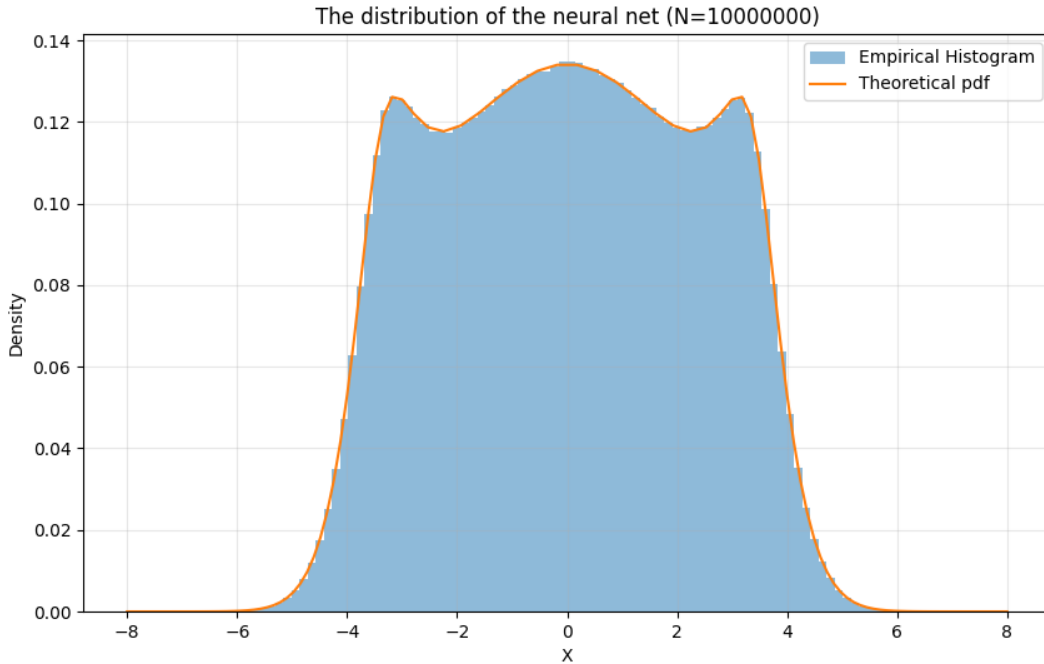


Figure 1: The distribution of the neural net for $\theta = [1.5, 0.4, 1.0, 0.2]$.

2 Tests

The first test I did was sensitivity analysis for variances to show the correctness of my gradient loss function. I started off with $\theta = [0.5, 0, 2.0, 1.0]$, since this gives us $X = Z$ so I know that the variance is 96 and I can compare it with my ssq value to see whether my loss function is correct. (It's correct since the ssq approaches the baseline variance as the

increment gets small). I used different increment sizes ([0.1, 0.01, 0.005, 0.002, 0.001]), and for the smallest increment, the gradient loss approaches the computed gradient loss value.

```
Computing Analytical Gradient (via gradient_loss function)
Starting from (theta = [0.5000, 0.0000, 2.0000, 1.0000])
baseline variance = 9.6065e+01, baseline loss = 1.0507e+02
computed sensitivity is ['-1.6803e+03', '3.5849e+02', '0.0000e+00', '0.0000e+00']
increments 1.0000e-01
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 4.7226e+01 | -9.7678e+02
b | 1.5156e+02 | 5.5491e+02
c | 9.6065e+01 | 0.0000e+00
d | 9.6065e+01 | 0.0000e+00
increments 1.0000e-02
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 8.8200e+01 | -1.5732e+03
b | 9.9782e+01 | 3.7165e+02
c | 9.6065e+01 | 0.0000e+00
d | 9.6065e+01 | 0.0000e+00
increments 5.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 9.2003e+01 | -1.6249e+03
b | 9.7889e+01 | 3.6478e+02
c | 9.6065e+01 | 0.0000e+00
d | 9.6065e+01 | 0.0000e+00
increments 2.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 9.4408e+01 | -1.6573e+03
b | 9.6787e+01 | 3.6077e+02
c | 9.6065e+01 | 0.0000e+00
d | 9.6065e+01 | 0.0000e+00
increments 1.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 9.5231e+01 | -1.6684e+03
b | 9.6425e+01 | 3.5945e+02
c | 9.6065e+01 | 0.0000e+00
d | 9.6065e+01 | 0.0000e+00
```

Figure 2: The sensitivity analysis for $\theta = [0.5, 0, 2.0, 1.0]$.

Then I repeated the analysis with a different theta value $\theta = [2.0, 0.45, 2.0, 0.15]$. I got the following results. Similarly, as the increment gets smaller, $\frac{dssq}{d\theta}$ approaches the gradient loss I computed from my function. This proves that my code successfully produces the correct gradient loss.

```

Computing Analytical Gradient (via gradient_loss function)
Starting from (theta = [2.0000, 0.4500, 2.0000, 0.1500])
baseline variance = 1.2417e+01, baseline loss = 2.1417e+01
computed sentivity is ['7.6028e+00', '-2.4989e-03', '7.4953e-03', '-5.9909e-02']
increments 1.0000e-01
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 1.3976e+01 | 7.7951e+00
b | 1.2417e+01 | -2.4998e-03
c | 1.2418e+01 | 4.1413e-03
d | 1.2416e+01 | -7.9858e-02
increments 1.0000e-02
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 1.2570e+01 | 7.6327e+00
b | 1.2417e+01 | -2.4998e-03
c | 1.2417e+01 | 7.0197e-03
d | 1.2417e+01 | -6.1740e-02
increments 5.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 1.2493e+01 | 7.6226e+00
b | 1.2417e+01 | -2.4998e-03
c | 1.2417e+01 | 7.2536e-03
d | 1.2417e+01 | -6.0832e-02
increments 2.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 1.2447e+01 | 7.6164e+00
b | 1.2417e+01 | -2.4998e-03
c | 1.2417e+01 | 7.3989e-03
d | 1.2417e+01 | -6.0292e-02
increments 1.0000e-03
parameters | ssq(theta0+dtheta) (computed) | dssq/dtheta
a | 1.2432e+01 | 7.6144e+00
b | 1.2417e+01 | -2.4998e-03
c | 1.2417e+01 | 7.4483e-03
d | 1.2417e+01 | -6.0113e-02

```

Figure 3: The sensitivity analysis for $\theta = [2.0, 0.45, 2.0, 0.15]$.

The second test I did was accessing which step size values ensures c_1 actual difference $\leq$ predicted difference $\leq c_2$ actual difference. The ratio (actual difference/predicted difference) is approximately 1.

Step Size	Pred Decrease	Actual Decrease	Ratio (Act/Pred)
5.00e-03	2.8860e-01	2.8757e-01	0.9964
1.00e-03	5.7719e-02	5.7755e-02	1.0006
5.00e-04	2.8860e-02	2.8892e-02	1.0011
1.00e-04	5.7719e-03	5.7807e-03	1.0015
1.00e-05	5.7719e-04	5.7813e-04	1.0016
1.00e-06	5.7719e-05	5.7813e-05	1.0016
1.00e-07	5.7719e-06	5.7813e-06	1.0016

Figure 4: The actual and predicted difference for $\theta = [0.5, 0, 2.0, 1.0]$.

3 New Results

I changed my step size to 0.0001. My batch size is still 1000. My algorithm gets me this graph, which has a smaller value around 0 but the wings' peak is around 6. I actually had an early stop to the result since The variance graph doesn't look right, as the variance keeps increasing, and a, b values are kept pushing up by the algorithm.

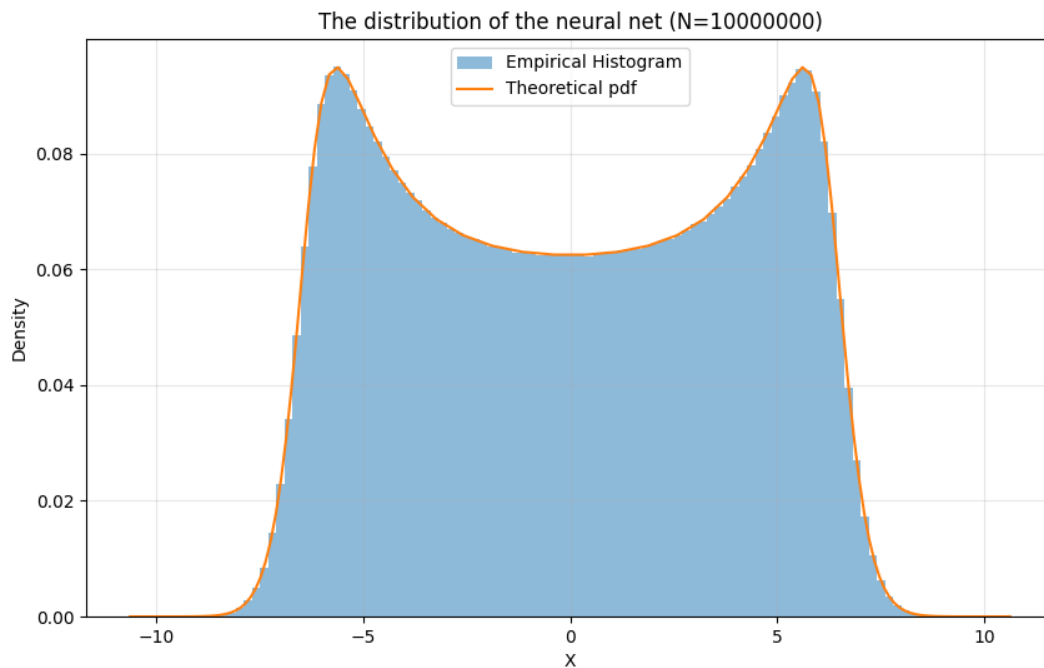


Figure 5: The distribution of the neural net for $\theta = [3.80, 2.02, 0.7, 0.31]$.

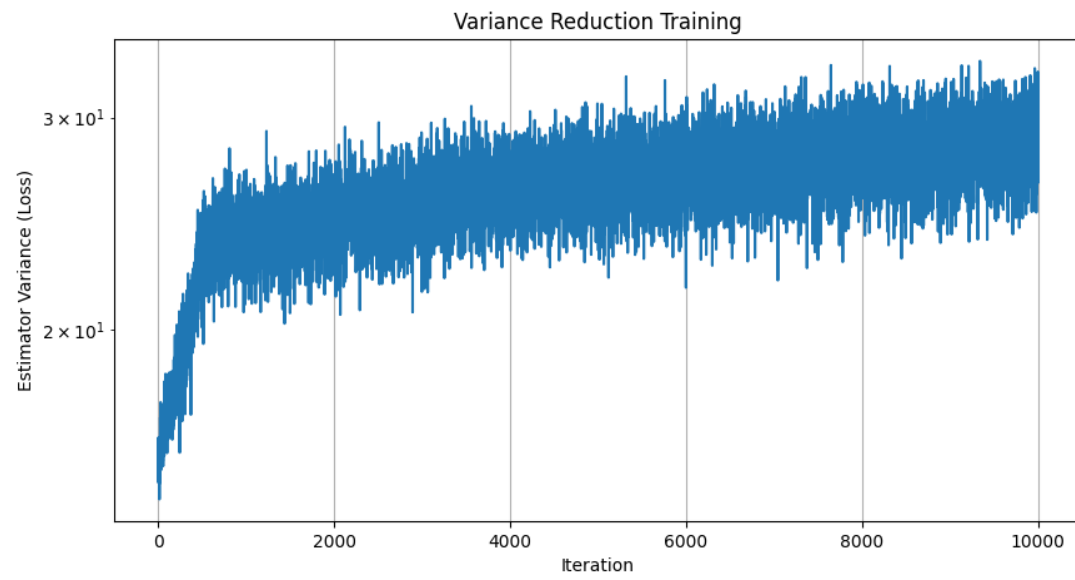


Figure 6: The variance change after $\theta = [1.5, 0.4, 1.0, 0.2]$.